Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-3(P1) | Date: 05-04-26
Time: 3 Hrs | 2022_P1 | Max. Marks: 180

KEY SHEET
MATHS

1	**265**	2	**2**	3	**5**	4	**4.55 or 4.56**	5	**5**
6	**10**	7	**995**	8	**0.16**	9	**BCD**	10	**AD**
11	**ACD**	12	**BC**	13	**AD**	14	**ABC**	15	**C**
16	**D**	17	**D**	18	**A**				

PHYSICS

19	**40.50**	20	**10**	21	**21.60**	22	**98**	23	**0.55 Or 0.56**
24	**3.10**	25	**40**	26	**450**	27	**ABCD**	28	**A**
29	**C**	30	**ABC**	31	**ABC**	32	**AC**	33	**A**
34	**B**	35	**D**	36	**A**				

CHEMISTRY

37	**35**	38	**900**	39	**4**	40	**1.60**	41	**7.50**
42	**9**	43	**2.25**	44	**1.25**	45	**ABC**	46	**BC**
47	**ABCD**	48	**ABC**	49	**B**	50	**ABD**	51	**C**
52	**B**	53	**D**	54	**B**				

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHS

1. $f(n)-nf(n-1)=-\left(f(n-1)-(n-1)f(n-2)\right)$

$\Rightarrow t_n=-t_{n-1},\ t_1=f(1)-f(0)=-1$

$\Rightarrow t_n=(-1)^n$

$\Rightarrow f(n)-nf(n-1)=(-1)^n \Rightarrow \dfrac{f(n)}{n!}-\dfrac{f(n-1)}{(n-1)!}=\dfrac{(-1)^n}{n!}$

Doing sum from $n=1$ to 6, $f(6)=265$

2. $(0,0,0),(1,1,1)$ are two trivial solutions.

Let $a>1\Rightarrow a^3>a^2$

$\Rightarrow b>a$

Similarly $c>b>a>1$ and $a=c^3-c^2+c>c \Rightarrow$ contradiction

If $a<1\Rightarrow a^3<a^2\Rightarrow b<a\Rightarrow c<b<a$

$\Rightarrow a<c<b<a \Rightarrow$ contradiction

3. $x^3+2026x=-1$

$y^3+2026y=-1$

$x=\tan p-1,\ y=1-\cot q$

$\Rightarrow (x-y)\left[x^2+y^2+xy+2026\right]=0\Rightarrow x=y$

$\Rightarrow \tan p+\cot q=2$

$\Rightarrow \sin r+\cos r=1$

Hence $r=0,\dfrac{\pi}{2},2\pi,-2\pi,\dfrac{-3\pi}{2}$

4. Let $\vec{a}=2\hat{i}+3\hat{j}+6\hat{k}$

$\vec{b}=(\sin x\sin y)\hat{i}+\cos y\,\hat{j}+(\cos x\sin y)\hat{k}$ and $\vec{a}.\vec{b}=|\vec{a}||\vec{b}|\Rightarrow \vec{a}\ \|\ \vec{b}$

$\dfrac{\sin x\sin y}{2}=\dfrac{\cos y}{3}=\dfrac{\cos x\sin y}{6}$

$\tan x=\dfrac{1}{3},\ \tan y=\dfrac{2}{3}\operatorname{cosec} x$

$=\dfrac{2}{3}\cdot\dfrac{\sqrt{10}}{1}$

$\tan^2 x+\tan^2 y=\dfrac{41}{9}=4.55$

5. $\lim\limits_{x\to 0}\dfrac{x^2\dfrac{\sin\beta x}{\beta x}\times\beta x}{\alpha n+\dfrac{(\alpha x)^3}{3}+\ldots-\left(x-\dfrac{x^3}{3}+\ldots\right)}=4$

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$$\Rightarrow \lim_{x\to 0}\frac{x^3\times\beta}{x(\alpha-1)+\left(\frac{\alpha^3}{3}+\frac{1}{3!}\right)x^3+....}=4$$

$$\Rightarrow \alpha=1 \text{ and } \frac{\beta}{\frac{1}{3}+\frac{1}{3!}}=4$$

$$\Rightarrow \beta=4\times\left(\frac{1}{3}+\frac{1}{6}\right)$$

$$=4\times\frac{1}{2}=2$$

6. $\left.\begin{matrix}a\sin b=1\\ a\cos b=-1\end{matrix}\right\}a=\pm\sqrt{2}$

$$\Rightarrow \text{If } a=\sqrt{2}\Rightarrow \sin b=-\cos b=\frac{1}{\sqrt{2}}$$

$$a=-\sqrt{2}\Rightarrow \sin b=-\cos b=-\frac{1}{\sqrt{2}}\Bigg\}10\,\text{pairs}$$

7. $$\frac{a+b}{2}=\sqrt{ab}+8$$

$$\Rightarrow a+b-2\sqrt{ab}=16$$

$$\Rightarrow \sqrt{b}-\sqrt{a}=4,\ 0<\sqrt{a},\sqrt{b}<10^3$$

So if

$$\sqrt{b}=999\Rightarrow\sqrt{a}=995$$

$$\sqrt{b}=998\Rightarrow\sqrt{a}=994$$

$$----------$$

$$----------$$

$$\sqrt{b}=5\Rightarrow\sqrt{a}=1$$

Therefore 995 such pairs are there

8. Let $z_1=e^{i\frac{2m\pi}{101}}$, $z_2=e^{i\frac{2k\pi}{101}}$

$$\left|e^{i\frac{2m\pi}{101}}+e^{i\frac{2k\pi}{101}}\right|\geq\sqrt{2+\sqrt{3}}$$

$$\Rightarrow\sqrt{\left(2+2\cos\frac{2\pi}{101}(m-k)\right)}\geq\sqrt{2+\sqrt{3}}$$

$$\Rightarrow\cos\left(\frac{2\pi(m-k)}{101}\right)\geq\frac{\sqrt{3}}{2}$$

$$\left|\frac{2\pi(m-k)}{101}\right|\leq\frac{\pi}{6}$$

$$|m-k|\leq\frac{101}{12}\Rightarrow|m-k|\leq 8$$

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$$P=\frac{101.16}{101.100}=\frac{4}{25}$$

9. $y=0,\ 3y+4x=0$ are asymptotes whose angle Bisector are

$y=2x,\ x+2y=0$

Solving H with $x+2y=0$

$$y^2=\frac{4}{5}\Rightarrow\pm\frac{2}{\sqrt{5}}$$

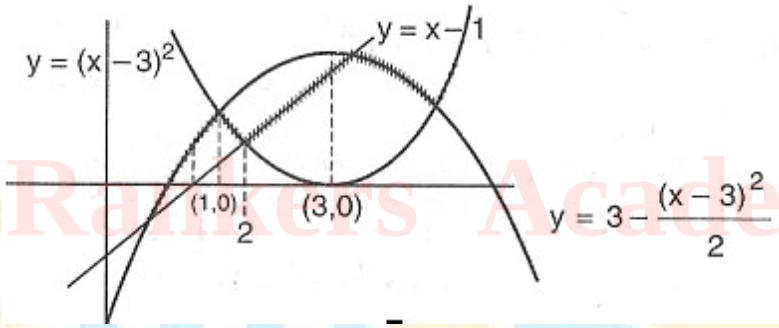
So vertex $\left(-\frac{4}{\sqrt{5}},\frac{2}{\sqrt{5}}\right)\left(\frac{4}{\sqrt{5}},-\frac{2}{\sqrt{5}}\right)$

$C.A: y=2x.\ a=2, b=1$

OR $H:\dfrac{\left(\frac{2x-4}{\sqrt{5}}\right)^2}{4}-\dfrac{\left(\frac{x+2y}{\sqrt{5}}\right)^2}{1}=1$

$$L=1,\ e^2=\frac{5}{4}$$

10. Clearly, $f(x)$ is always continuous and non-derivable at two points in $(1,3)$.



11. $\because\ x=\sqrt{5+\sqrt{+\sqrt{5+.....}}}\ \Rightarrow\ x=\sqrt{5+x}$

$\Rightarrow x^2-x-5=0$

$\therefore P(x)=x^2-x-5$

$$\therefore\frac{1}{P(r)+5}=\frac{1}{r^2-r}=\frac{1}{r-1}-\frac{1}{r}$$

$$100\sum_{r=2}^{100}\left(\frac{1}{r-1}-\frac{1}{r}\right)=100\left(1-\frac{1}{100}\right)=99$$

$$\frac{1}{P(r)+3}=\frac{1}{r^2-r-2}=\frac{1}{(r-2)}-\frac{1}{r+1}$$

$$\therefore\lim_{n\to\infty}\sum_{r=3}^{n}\left(\frac{1}{r-2}-\frac{1}{r+1}\right)=1+\frac{1}{2}+\frac{1}{3}=\frac{11}{6}$$

and $\displaystyle\int_0^1\frac{1}{P(x)+x+6}dx=\int_0^1\frac{1}{x^2+1}dx=\left(\tan^{-1}x\right)_0^1=\frac{\pi}{4}$

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and $\int_1^2 \frac{1}{\sqrt{P(x)+x+4}}dx = \int_1^2 \frac{1}{\sqrt{x^2+1}}dx = \left(\ln\left(x+\sqrt{x^2-1}\right)\right)_1^2 = \ln\left(2+\sqrt{3}\right)$

12. $\because\ R_n = \sum_{r=1}^{2n} x^{4r-3} = x + x^5 + x^9 .....$

$S_n = \sum_{r=1}^{2n} x^{4r-2} = x^2 + x^6 + x^{10} .....$

$T_n = \sum_{r=1}^{2n} x^{4r-1} = x^3 + x^7 + x^{11} .....$

$U_n = \sum_{r=1}^{2n} x^{4-r} = x^4 + x^8 + x^{12} .....$

Clearly, $R_n > S_n > T_n > U_n$ as $x \in \left(0, \frac{\pi}{4}\right)$

and $\lim_{n\to\infty}\left(R_n + S_n + T_n + U_n\right) = x + x^2 + ... = \frac{x}{1-x}$

$\because\ R_n + S_n = T_n + U_n$

$\Rightarrow \frac{x\left(1-x^{8n}\right)}{1-x^4} + \frac{x^2\left(1-x^{8n}\right)}{1-x^4} = \frac{x^3\left(1-x^{8n}\right)}{1-x^4} + \frac{x^4\left(1-x^{8n}\right)}{1-x^4}$

$\Rightarrow x + x^2 = x^3 + x^4 \Rightarrow \left(x+x^2\right)\left(x^2-1\right) = 0 \Rightarrow$ no value of x.

13. Clearly plane P will be perpendicular bisector of line joining A and B.

$\therefore$ It passes though $(-1, -1, 2)$ and drs of its normal are $(3, -3, 1)$

$\therefore$ Equation of plane of P is

$3(x+1) - 3(y+1) + 1(z-2) = 0$

$\Rightarrow\quad 3x - 3y + z = 2$

(a) $\because$ P is parallel to Q

$\therefore\quad \frac{1}{3} = \frac{1}{3} = \frac{C}{1} \Rightarrow C = \frac{1}{3}$

(b) If P is perpendicular to Q then

$1\cdot 3 = 1\cdot(-3) + 1\cdot C = 0 \Rightarrow C = 6$

(c) $\because$ Plane P intersects coordinate axes at points with position vectors $\frac{2}{3}\hat{i}, -\frac{2}{3}\hat{j}$ and $2\hat{k}$

$\therefore\quad$ Volume of tetrahedron $= \frac{1}{6}\left|\left[\frac{2}{3}\hat{i} \quad -\frac{2}{3}\hat{j} \quad 2\hat{k}\right]\right| = \frac{4}{27}$

(d) Point on line L can be taken as $(\lambda+1, 3\lambda-2, -\lambda+7)$

$\therefore\quad$ It is on plane P.

$\therefore\quad 3(\lambda+1) - 3(3\lambda-2) + (-\lambda+7) = 2$

$\Rightarrow\quad 7\lambda = 14 \Rightarrow \lambda = 2$

$\therefore\quad$ point of int. $= (3, 4, 5)$

$\therefore\quad \alpha + \beta + \gamma = 12$

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14. $A^2=\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}=\begin{bmatrix}9&8&0\\8&9&8\\8&8&9\end{bmatrix}$;

Also $\det A=1(1-4)-2(2-4)+2(4-2)$

$=-3+4+4=5$

$\therefore$ $\det A\neq 0 \Rightarrow A$ is invertible

We have $A^2-4A-5I_3$

$=\begin{bmatrix}9&8&8\\8&9&8\\8&8&9\end{bmatrix}-4\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}-5\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=O$

$\Rightarrow 5I_3=A^2-4A=A(A-4I_3)$

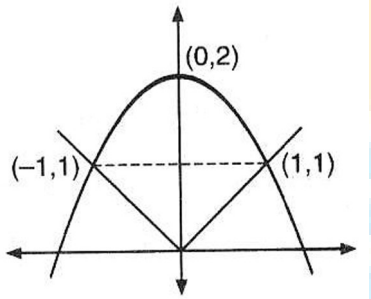
$\Rightarrow I_3=A\left[\frac{1}{5}(A-4I_3)\right]\Rightarrow A^{-1}=\frac{1}{5}(A-4I_3)$

15. a) CS_1 and CS_2 divide the triangle into three of equal area so that S_1 and S_2 are points of trisection of AB. Thus $S_1=(0,0)$, $S_2=(2,-3)$. Equation of CS_1 and $y-x=0$. Slope of $CS_2=-4$. The line through $(0,0)$ drawn parallel to CS_2 is $y+4x=0$

(b) The required area is bounded by the line $y=\pm x$ and the parabola $x^2=-(y-2)$ having the vertex at $(0,2)$ and passing through $(\pm 1,1)$. By symmetry about the y-axis, Area

$=2\left[\frac{1}{2}+\int_1^2\sqrt{2-y}\,dy\right]=\frac{7}{3}\equiv\frac{p}{q}$

$\Rightarrow p-q=7-3=4$



(c) With $y=xv$, $xv'=\frac{1+v^2}{2v}-v=\frac{1-v^2}{2v}\Rightarrow\frac{2v\,dv}{1-v^2}=\frac{dx}{x}$

Whose solution is $x(1-v^2)=c$. When $x=2$, $v=\frac{1}{2}$ so that $c=\frac{3}{2}$.

The equation of the curve is $x^2-y^2=\frac{3x}{2}\Rightarrow\left(x-\frac{3}{4}\right)^2-y^2=\frac{9}{16}$

This is a rectangular hyperbola with eccentricity $\sqrt{2}$.

$\therefore\ e^4=4$

(d) $P=\lim_{n\to\infty}\frac{\left[(n^2+1^2)(n^2+2^2)....(n^2+n^2)\right]^{m/n}}{n^{2m}}$

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$$\ln P = \lim_{n\to\infty} \frac{m}{n}\left[\sum_{r=1}^{n} \ln\left(n^2+r^2\right) - 2m\ln n\right]$$

$$\ln P = m\lim_{n\to\infty}\left[\sum_{r=1}^{n} \frac{\ln\left(n^2+r^2\right)-2n\ln n}{n}\right]$$

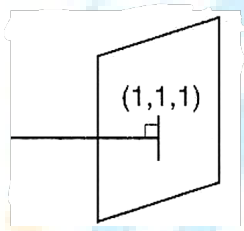
$$\ln P = m\left[\ln 2 + \frac{\pi}{2} - 2\right]$$

$$P = e^{m\left[\ln 2+\frac{\pi}{2}-2\right]} = e^{m\ln 2}\cdot e^{-2m}\cdot e^{\frac{m\pi}{2}} = \frac{2^m e^{\frac{m\pi}{2}}}{e^{2m}} = \frac{\left(2e^{\pi/2}\right)^m}{e^{2m}}$$

$$= \left(\frac{2\sqrt{e^{\pi}}}{e^2}\right)^m \Rightarrow \alpha = 2$$

16. (I) Equation of the plane $A(x-1)+B(y-1)+C(z-1)=0$ .....(1)

Since the line is perpendicular to the plane (1)



$\therefore \quad 3(x-1)+0(y-1)+4(z-1)=0$

$3x+0y+4z-7=0$

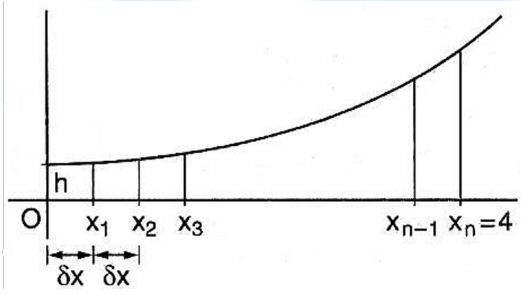
Distance from $(0, 0, 0)$

$$d = \left|\frac{-7}{5}\right| = \frac{7}{5} = \frac{p}{q}$$

$\Rightarrow \quad p-q=2$

(II) $L = \lim_{\delta x\to 0} \delta x\left(x_1+x_2+x_3+....+x_n\right)$

$$= \lim_{n\to\infty}\frac{4}{n}\left[\frac{4}{n}+\frac{8}{n}+\frac{12}{n}+.....+4\cdot\frac{n}{n}\right]\left(\delta x = \frac{4}{n}\right)$$



$$= \lim_{n\to\infty}\frac{16}{n^2}(1+2+3+.....+n) = \lim_{n\to\infty}\frac{16}{n^2}\cdot\frac{n(n+1)}{2} = 8$$

(III) If x is replaced by $-x$ in the given equation then

$-x f(-x)+(1+x)f(x) = x^2-x+1$

subtracting the two equations we get

$f(-x) = f(x)+2x,$

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substituting the value of $f(-x)$ in the given equation we get

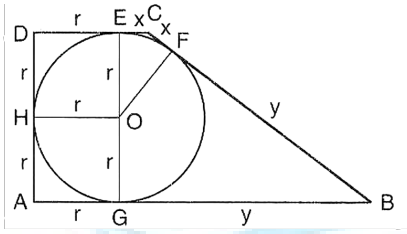
$x f(x)+(1-x)(f(x)+2x)=x^2+x+1$ and thus

$$f(x)=x^2+x+1-2x(1-x)=3x^2-x+1=3\left(x-\frac{1}{6}\right)^2+\frac{11}{12}$$

and hence $f(x)\geq\frac{11}{12}$. If $x=\frac{1}{6}$ then we get $f\left(\frac{1}{6}\right)=\frac{11}{12}\equiv\frac{p}{q}$

$\Rightarrow (q-p)=1$

(d)



Using the notations from the above diagram and the conditions from the problem one obtains:

$3(r+x)=r+y \Leftrightarrow y=3x+2r$

and $(x+y)^2=4r^2+\{(r+y)-(r+x)\}^2 \Leftrightarrow xy=r^2$

Hence $x(3x+2r)=r^2 \Leftrightarrow x=\frac{r}{3}$

and consequently $y=3r$. On the other hand the area of the trapezoid ABCD is 4, thus $(r+x+r+y)=4$

Substituting of $x=\frac{r}{3}$ and $y=3r$, we get $r=\frac{\sqrt{3}}{2}\Rightarrow 4r^2=3$

17. (a) $x+2x+.....12x=78x$

$\therefore \{x\}=x\cdot\{2x\}=2x.....\{12x\}=12x$

$\therefore 0\leq 12x<1 \Rightarrow 0\leq x<\frac{1}{12}$

Since $x\in\left[\frac{1}{15},\frac{1}{10}\right]$

$\therefore x\in\left[\frac{1}{15},\frac{1}{12}\right)\Rightarrow x=\frac{1}{15},\frac{1}{14},\frac{1}{13}$

$\therefore$ number of values of $x=3$

(b) $\because \left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20}a_r x^r$ replace x by $\frac{2}{x}$ we get

$$\left(\frac{8}{x^2}+\frac{6}{x}+4\right)=\sum_{r=0}^{20}a_r\frac{2^r}{x^r}$$

$$\Rightarrow 2^{10}\left(4+3x+2x^2\right)^{10}=\sum_{r=0}^{20}a_r 2^r x^{20-r}$$

Comparing coefficient of x^9 both sides, we get

$$2^{10}\cdot a_9=2^{11}a_{11}\Rightarrow\frac{a_9}{a_{11}}=2$$

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(c) Let P be origin and position vectors of point A, B, C be $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$

$\because \left|\vec{\alpha}, \vec{\beta}, \vec{\gamma}\right|^2 + \left|\vec{\alpha}+\vec{\beta}\right|^2 + \left|\vec{\beta}+\vec{\gamma}\right|^2 + \left|\vec{\gamma}-\vec{\alpha}\right|^2 = 3\left(\left|\vec{\alpha}\right|^2 + \left|\vec{\beta}\right|^2 + \left|\vec{\gamma}\right|^2\right)$

$\Rightarrow 9PG^2 + AB^2 + BC^2 + CA^2 = 3\left(PA^2 + PB^2 + PC^2\right)$

$\Rightarrow 9\times 1 + a^2 + b^2 + c^2 = 3(1+4+1)$

$\Rightarrow a^2 + b^2 + c^2 = 9$

(d) $\because f(xy) = f(x) + f(y)$

$\Rightarrow f(16) = f(4) + f(4) = 3 \Rightarrow f(4) = \frac{3}{2}$

$\Rightarrow f(2) = f(2) = \frac{3}{2} \Rightarrow f(2) = \frac{3}{4}$

$\therefore \frac{1}{f(2)} + \frac{1}{f(4)} = \frac{4}{3} + \frac{2}{3} = 2$

18. (a) $25^{24^{26}} \equiv 3^{24^{26}} \equiv 3^{6\lambda} \equiv 1 \pmod 7$

(b) $\frac{x^8 + y^8 + 1+1+1+1+1+1}{8} \geq \left(x^8 \cdot y^8\right)^{\frac{1}{8}} \Rightarrow (x, y) = (1, 1) \text{ or } (-1, -1)$

(c) Image (h, k) of focus $(-1, -1)$ in tangent $3x - y - 8 = 0$ will lie on directrix.

$\frac{h+1}{3} = \frac{k+1}{-1} = \frac{-2(3(-1) - 1(-1) - 8)}{3^2 + 1^2} = 2$

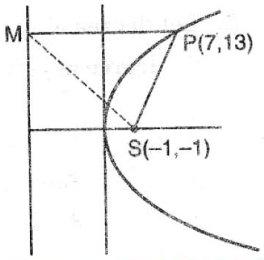
$\therefore$ Image $(5, -8)$ i.e., on directrix

$\therefore$ Slope of $PM = \frac{13+3}{7-5} = 8$

$\therefore$ Equation of directrix will be $y + 3 = \frac{-1}{8}(x - 5)$

$\Rightarrow x + 8y + 19 = 0$

$\therefore [l] = 2$



(d) Distance between centre and focus = ae = 10

Distance between directrices $= \frac{2a}{e} = 4$

$\therefore \frac{ae}{2a} = \frac{10}{4} \Rightarrow e^2 = 5$

$\Rightarrow \frac{4e^2}{5} = 4$

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PHYSICS

24. The bead will move along the elliptical path in vertical plane.
Such that $2a = 10\ell$ and $2ae = 8\ell \Rightarrow a = 5\ell,\ b = 3\ell$
Let be radius of curvature of the path followed by the bead initially.
4Equation of ellipse is

$$\frac{x^2}{(5\ell)^2} + \frac{y^2}{(3\ell)^2} = 1$$

$$r_0 = \frac{\left[1+(y')^2\right]^{3/2}}{|y''|} \text{ at } (0,-3\ell) \left(\because y' = 0, y'' = \frac{3}{25\ell}\right)$$

$$\therefore\ r_0 = \frac{25\ell}{3}$$

$$2\ T_0\cos\theta - mg = \frac{m}{r_0}(\sqrt{2\ g\ell})^2 \left(\because \cos\theta = \frac{3}{5}\right)$$

$$T_0 = \frac{31mg}{30}.$$

26. Number of beats heard in a minute is beat frequency in *Hz* times 60.

28. Magnetic field is independent of speed of the sheet as the current due to movement of sheet is equal to rate of deposition of charge which is constant

29. $\cos\theta \frac{d\theta}{dt} = \frac{-v}{l}$

33. $5\times10^{-15} = eV_0$

$$V_0 = \frac{5\times10^{-15}}{1.6\times10^{-19}} = 3.125\times10^4 \text{ V}$$

$$\Rightarrow \Delta E = 5\times10^{-15} - 6\times10^{16} = 4.4\times10^{-15} \text{ J}$$

$$\Rightarrow 4.4\times10^{-15} = \frac{hC}{\lambda}$$

$$\lambda = \frac{6.67\times10^{-34}\times3\times10^8}{4.4\times10^{-15}} = 4.44\times10^{-11} \text{ m}$$

$\Rightarrow$ Kinetic energy of electron emitted from M-level

$4.4\times10^{-15} - 7\times10^{-17} = 4.33\times10^{-15}$ J

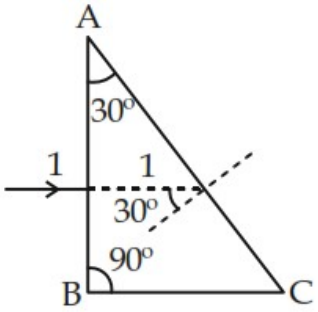
34. (P) Refractive index of the prism is the minimum value required for ray (1) to undergo total internal reflection at face AC . Ray (1) falls on face AC at an angle of incidence 30°

$$\therefore 30^\circ > i_C$$

$$\sin30^\circ > \sin i_C$$

$$\therefore \mu > 2$$

Minimum value of μ can be taken as 2.



(Q) For ray 2, refractive angle of prism is 30°. Apply Snell's law for refraction at face AB.

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$1\sin i = \mu \sin r$

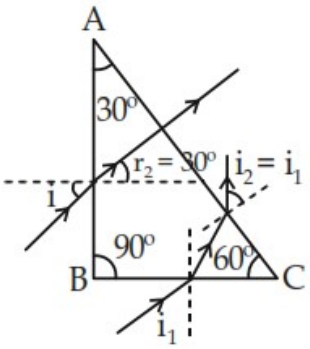
$i = 90^\circ$

(R) Using the relation $i_1 + i_2 = A + \delta$ for ray 2 .

$90^\circ + 0^\circ = 30^\circ + \delta$

$\delta = 60^\circ$

(S) $\mu = \dfrac{\sin\left(\dfrac{A+\delta m}{2}\right)}{\sin\dfrac{A}{2}} \Rightarrow \delta m = 120^\circ$



35. $V^2 = 144 - 9x^2$

$\omega^2 = 9$

$\Rightarrow \dfrac{2\pi}{T} = 3 \Rightarrow T = \dfrac{2\pi}{3}$

when $v = 0, x = \sqrt{\dfrac{144}{9}} = \dfrac{12}{3} = 4$ unit

Acceleration $= v\dfrac{dv}{dx} = -9x$

Potential energy $= \dfrac{1}{2}m\omega^2 x^2 = 18$ units

36. Length of solenoid, $l = 1.20$ m

Diameter of solenoid, $d = 2.50$ cm $\Rightarrow r = 1.25\times10^{-2}$ m

Number of turns, $N = 1400$

Current, $I = 2.00$ A

Diameter of superconducting bar, $d_{bar} = 2.20$ cm $\Rightarrow r_{bar} = 1.10\times10^{-2}$ m

Superconductor shows perfect diamagnetism (expels magnetic field).

(A) Magnetic field inside the solenoid:

$$B = \mu_0 nI = \mu_0 \frac{N}{\ell} I = \left(4\pi\times10^{-7}\right)\cdot\frac{1400}{1.2}\cdot 2 = 2.93\times10^{-3}\ T = 2.93 mT$$

(B) Energy density of magnetic field:

$$U = \frac{B^2}{2\mu_0} = \frac{\left(2.93\times10^{-3}\right)^2}{2.4\pi\times10^{-7}} = \frac{8.58\times10^{-6}}{8\pi\times10^{-7}} \approx 3.42\ Pa$$

(C) Force on the superconducting bar: Because the field is expelled from the superconductor, the bar experiences a force pushing it out of the field region. This force is:

$$F = u\cdot A = \frac{B^2}{2\mu_0}\cdot\pi r^2 = 3.42\cdot\pi\cdot\left(1.1\times10^{-2}\right)^2 = 3.42\cdot\pi\cdot1.21\times10^{-4} \approx 1.30\times10^{-3}\ N = 1.30\ mN$$

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CHEMISTRY

37. $Co^{2+} + 6NH_3 \rightleftharpoons \left[Co(NH_3)_6\right]^{2+} \quad K_1 = 10^5$

$Co^{3+} + 6NH_3 \rightleftharpoons \left[Co(NH_3)_6\right]^{3+} \quad K_2 = 10^x$

$Co^{3+} + e \rightarrow Co^{2+} \quad E^\circ = 1.9V$

$$\frac{\left[Co(NH_3)_6\right]^{2+} \rightarrow \left[Co(NH_3)_6\right]^{3+} + e \quad E^\circ = -0.1\,V}{Co^{3+} + \left[Co(NH_3)_6\right]^{2+} \rightleftharpoons Co^{2+} + \left[Co(NH_3)_6\right]^{3+}} \quad E^\circ = 1.8V$$

$$E^\circ = \frac{0.06}{1}\log\frac{K_2}{K_1} \text{ (at equilibrium)}$$

$$1.8 = 0.06 \log \frac{K_2}{10^5}$$

$$30 = \log\frac{K_2}{10^5}$$

$$K_2 = 10^{35}$$

38. $$\log\frac{k_2}{k_1} = \frac{30.8 \times 298 \times 4.606}{2.303 \times 2} \times \left[\frac{308-208}{308 \times 298}\right]$$

$$\log\frac{k_2}{k_1} = 1$$

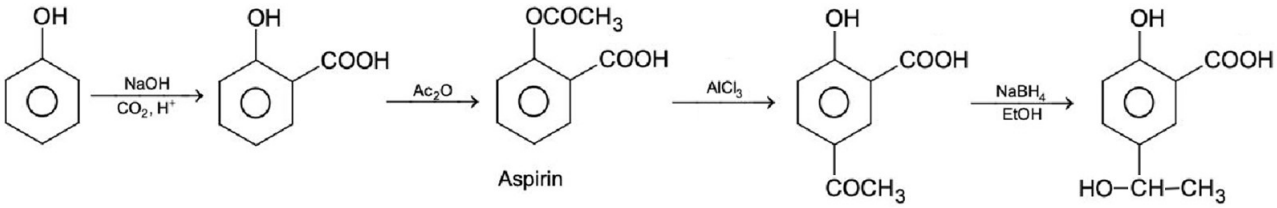
$$\frac{k_2}{k_1} = 10$$

$$k_2 = 10k_1$$

$$\% \text{ increased } (x) = \frac{10k_1 - k_1}{k_1} \times 100$$

$$= 900.00$$

39.



40. $pH = 12,\ pOH = 2$

$\left[OH^-\right] = 10^{-2}$

Equivalents of $\left[OH^-\right] = 10^{-2} \times 80 \times 10^{-3} = 8 \times 10^{-4}$

$$8 \times 10^{-4} = \frac{9.65 \times x}{96500}$$

$x = 8$ sec

$$\text{Equivalents of oxygen} = \frac{0.28}{5.6} = 0.05$$

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$$0.05 = \frac{965 \times y}{96500}$$

$$y = 5$$

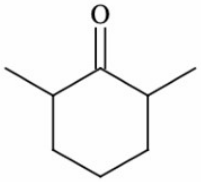
41. This is a buffer solution where $H_2PO_4^-$ acts as the acid and NaH_2PO_4 is the salt.

$$\therefore pH = pK_2 + \log \frac{\left[HPO_4^-\right]}{\left[H_2PO_4^-\right]}$$

$$\Rightarrow 7.2 + \log 2$$

$$\Rightarrow 7.50$$

42. (I) 5

Except $C_2H_5-\overset{O}{\overset{\|}{C}}-CH_3$, $CH_3-\overset{O}{\overset{\|}{C}}-H$, , 

(II) 4

43. 6 compounds has 1 mole each

NH_4NO_3 do not give O_2

$$\underset{(1\text{ mole})}{2LiNO_3} \rightarrow Li_2O + 2NO_2 + \frac{1}{2}\underset{(0.25\text{ mole})}{O_2}$$

$$\underset{(1\text{ mole})}{2Ca(NO_3)_2} \rightarrow 2CaO + 4NO_2 + \underset{(0.5\text{ mole})}{O_2}$$

$$\underset{(1\text{ mole})}{2Pb(NO_3)_2} \rightarrow 2PbO + 4NO_2 + \underset{(0.5\text{ mole})}{O_2}$$

$$\underset{(1\text{ mole})}{2AgNO_3} \rightarrow 2Ag + 2NO_2 + \underset{(0.5\text{ mole})}{O_2}$$

$$\underset{(1\text{ mole})}{NaNO_3} \rightarrow NaNO_2 + \frac{1}{2}\underset{(0.5\text{ mole})}{O_2}$$

Total number of moles of O_2 evolved = 2.25

44. Number of atoms of X in FCC packing (at corners and face centres of cubic unit cell)

$$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

Number of atoms of Y at octahedral voids = 4

Along one body diagonal there are two X atoms and one Y-atom.

No. of effective atoms of X after removal $= 4 - 2 \times \frac{1}{8} = \frac{15}{4}$

No. of atoms of Y after removal = $4 - 1 = 3$

$$x : y = \frac{15}{4} : 3$$

$$= 5 : 4$$

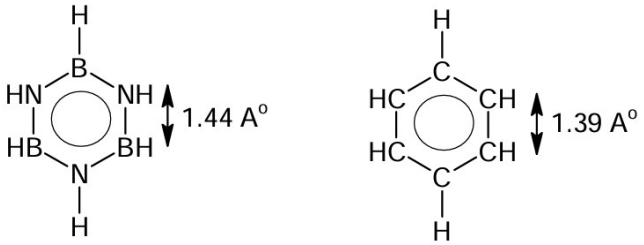
Simplest formula $= X_5Y_4$

$a = 5, b = 4, a/b = 1.25$

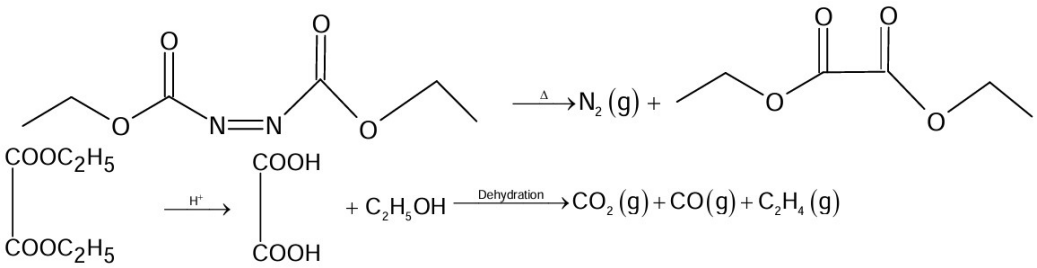
45. Only (D) option is correct. π electron cloud of $C_2 - C_3$ is present in same plane.

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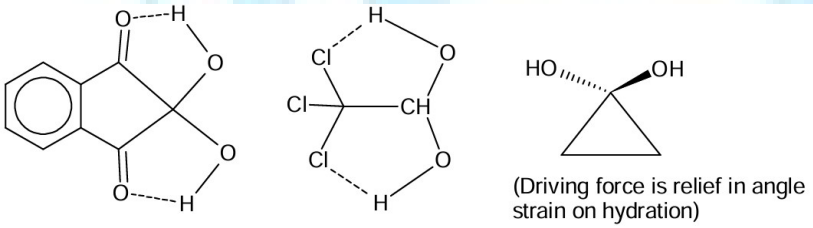
46. Organic benzene being a more symmetrical molecule has higher melting point $(6°C)$ than inorganic benzene (melting point $= -58°C$)



47.



48.

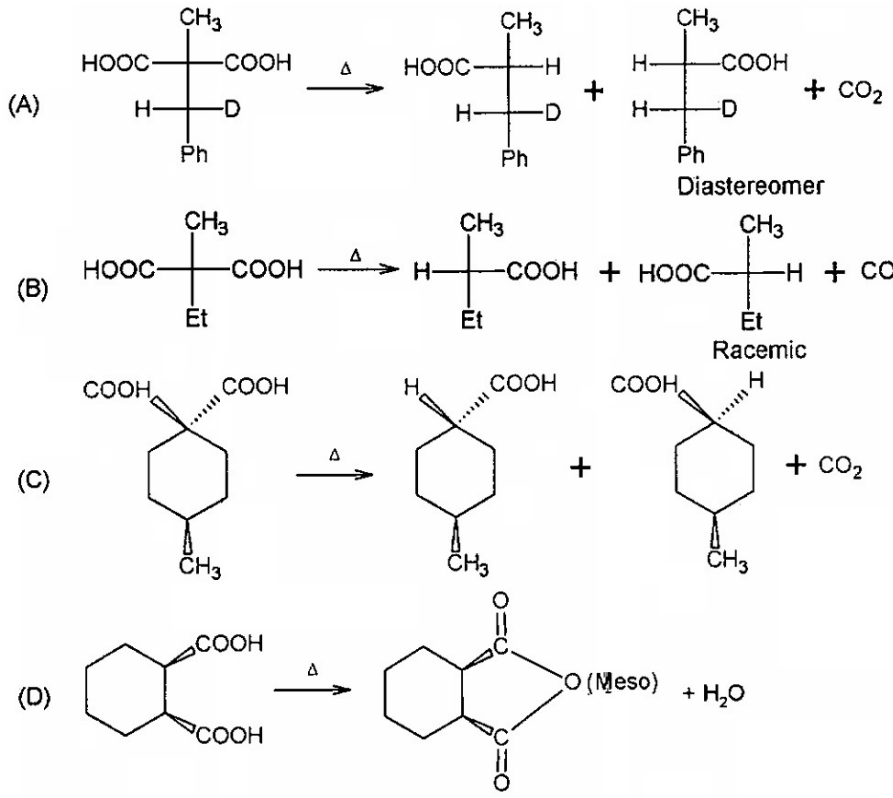


49. $2Na_2CrO_4 + H_2O + 2CO_2 \rightarrow Na_2Cr_2O_7 + 2NaHCO_3$

$3K_2MnO_4 + 2H_2O + 4CO_2 \rightarrow 2KMnO_4 + MnO_2 + 4KHCO_3$

50. Na / C_2H_5OH can reduce acid.

51.

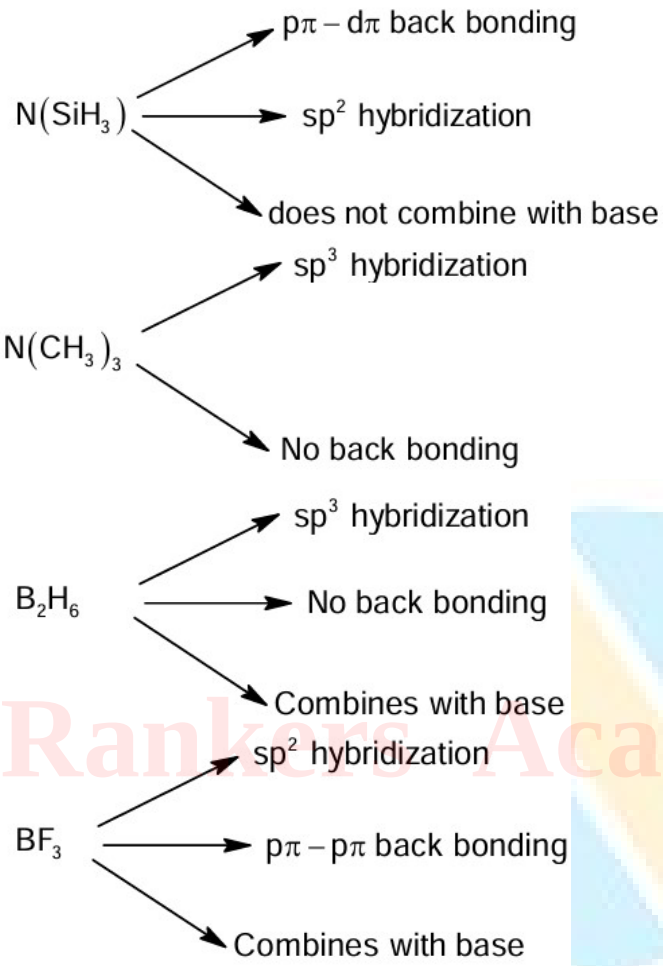


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52.

$[Ma_3b_2c]$	Geometrical isomerism $\rightarrow 3$	Stereoisomer $\rightarrow 3$
$[Ma_3b_3]$	Geometrical isomerism $\rightarrow 2$	Stereoisomer $\rightarrow 2$
$[Ma_3bcd]$	Geometrical isomerism $\rightarrow 4$	Stereoisomer $\rightarrow 5$
$[Ma_4bc]$	Geometrical isomerism $\rightarrow 2$	Stereoisomer $\rightarrow 2$

53.



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